ECON10005 Quantitative Methods 1

Tutorial in week 7: Week 6 Material

Zheng Fan

The University of Melbourne

Introduction

Zheng Fan

- Ph.D student in Economics at Unimelb.
- Personal website: *zhengfan.site* for some details

Don't be shy if you need help

- Discuss on Ed Discussion Board
- Attend consultation sessions: see Canvas for time and location
- Consult Stop 1, in case of special considerations,
- Contact QM-1@unimelb.edu.au for admin issues
- Send me an email: fan.z@unimelb.edu.au

Agenda

Today's focus

- Understanding sampling distributions
- Central Limit Theorem (CLT)
- Practice with normal probability calculations

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- Overall, you have made a great start—well done to everyone!
- This is your first report; the primary objective is to learn and refine your skills.
- Losing marks here is expected and useful. It highlights exactly where to focus your attention for the next submission.
- Rather than being discouraged by lost points (just 3%), use the feedback to improve your 7%.

PART A Questions

Answer the following questions.

1. Define a sampling distribution in your own words. How does it differ from a population distribution?
2. Explain how the mean and standard deviation of a sampling distribution are related to the population parameters. What happens to the standard deviation of the sampling distribution as the sample size increases?
3. State the Central Limit Theorem (CLT). Why is the CLT important for inferential statistics, especially in business decision-making?

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- An example of a statistic is the sample mean $\bar{X}$ or the sample proportion $\hat{p}$
- The distribution shows all possible statistics values that could be obtained from all possible samples of a certain size taken from the population.

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- $\hat{p}$ can be $0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}, 1$ where n is the sample size.

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- i is the index for each sample, $1, 2, \ldots, n$.

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- A sampling distribution shows how sample statistics (like the mean income $\bar{X}$ of randomly selected groups) would distribute across many samples X_i
- Recall that $\bar{X} = \dfrac{X_1 + \cdots + X_n}{n}$

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- As sample size n increases, SE decreases $\rightarrow$ more precise estimates

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- Allows use of normal distribution for inference.
- Critical for applications in business where underlying data is not normal.

PART B Questions

A tutor collects data from **10 students** in their own tutorial and finds an average MST mark of **16.5 / 20**.

The Tutor's Claim: *"Students in QM1 are performing very well on the MST."*

In your pre-tutorial Part B work, you were asked 7 questions about the above scenario. Let's spend some time in pairs or small groups discussing the answers you found before we go through the answers as a class.

The Tutor's Conclusion

Scenario: A tutor collects data from **10 students** in their own tutorial and finds an average MST mark of **16.5 / 20**.

The Tutor's Claim: *"Students in QM1 are performing very well on the MST."*

Discussion Questions:

1. Do you agree with this conclusion?
2. Does this sample represent all 1,400 QM1 students?

1. Validity of the Conclusion

Do we agree? Not necessarily.

- **Small Sample Size:** $n = 10$ is very small relative to $N = 1,400$.
- **Scope:** The data is restricted to one tutorial group only.
- **Generalizability:** A mean of 16.5 suggests this specific group is doing well, but is insufficient evidence for the entire cohort.

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2. Is it Representative?

- **Unlikely.** Tutorials often differ in student ability, engagement, or even the time of day (e.g., 8 AM vs. 4 PM).
- **Selection Bias:** Because the sample isn't randomly selected from the whole population, it likely reflects a specific subset, not the cohort.

3. Sampling Variability

Question: If different tutors take their own samples of 20 students, will they get the same mean?

Answer: No.

- **Sampling Variability:** Statistics calculated from different samples naturally vary.
- **Systematic Differences:** Because these aren't Simple Random Samples), means may differ due to tutorial-specific factors (e.g., a "high-performing" tutorial).

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4. What would you expect most sample means to be?

- Most sample means would be **close to 16.5**, but almost never exactly 16.5.
- **Warning:** If the sampling method is biased, these means might be systematically higher or lower than the *true* population mean.

5. Factors Affecting Variability

What factors affect how much the sample means varies?

- **Sample Size (n):** Larger samples reduce variability and increase precision.
- **Sampling Method:** Random sampling reduces bias and ensures representativeness.
- **Population Variability (σ):** If student marks are widely spread out, the sample means will also show more variation.

6. Identifying the Problem

Identify the problem with the tutor's sampling method and the consequences.

The Flaw: Convenience Sampling The tutor sampled based on ease of access, not random selection.

Consequences:

- **Bias:** The results are likely systematically skewed.
- **Unreliability:** Conclusions about the 1,400 students are misleading.

7. How to Improve the Study

How could the tutors improve?

A. Improve the Method

- Use **Random Sampling** from the entire 1,400 student list.
- Ensures every student has an equal and independent chance of selection.

B. Improve the Sample Size

- Increase n beyond 10 or 20.
- Makes the estimate more "stable" and likely to be closer to the true population mean.

Summary: Characteristics of a "Good" Sample

For statistical inference to be valid, a sample should be:

Representative Reflects key population characteristics so results can be generalized.

Random Every member has an equal chance of selection; reduces bias.

Independent Selection of one student doesn't influence the selection of another.

Sufficiently Large Reduces variability and improves precision/reliability.

Any final questions?

Thanks for your attention! 😄

Let me know if you have any questions.